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CONCEPTS User's Manual

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For use beginning with
TopAGNPS v6.00

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CONCEPTS User's Manual:

Introduction:

“CONCEPTS” is a module in the suite of modules called TopAGNPS. Its purpose is to allow the user to create a CONCEPTS corridor based on a user-specified upstream and downstream reach ID. The corridor is defined as the stream network of all the connecting reaches between and including the user-specified upstream and downstream reach IDs and all contributing cells to those reaches.

Most of the input files for CONCEPTS, as described below, can be generated by executing the following modules: DEDNM, RASPRO, RASFOR, and AgFlow. The TopAGNPS modules, (DEDNM, RASPRO, and RASFOR), produces spatial files required by CONCEPTS. AgFlow produces spatial output files called “AgFlow_Reach_Data.csv”, “AnnAGNPS_Cell_IDs.asc” and “AnnAGNPS_Reach_IDs.asc” which are also required by CONCEPTS.

CONCEPTS produces several output files which are described below.

Input files:

There are five input files, described below, necessary for the successful execution of CONCEPTS. The first is a control file and the remaining four are spatial layer files. The layer files are specifically named as described below.

Control file:

CONCEPTS.CSV or CONCEPTS.XML

The required control file allows the user to specify a number of parameters that affect the processing of spatial data and the determination of CONCEPTS corridors. If both “CONCEPTS.CSV” and “CONCEPTS.XML” are found, the “CONCEPTS.CSV” control file will be processed. A message will be produced in the log file and CONCEPTS execution terminated if the control file is not present.

“CONCEPTS.CSV” is a csv-formatted control file. The first record in the control file is the header record containing four possible keywords that may be used. The second record is the data record containing the values for the keywords that are specified in the header record. Only two records are read, header and data; therefore, any other information contained within the control file is ignored. The control file must be named “CONCEPTS.CSV”.

The following is an Excel example of the csv-formatted control file showing all of the available keywords:

UPSTREAM_REACH_ID	DOWNSTREAM_REACH_ID	ASC_PATH	CSV_PATH
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UPSTREAM_REACH_ID	DOWNSTREAM_REACH_ID	ASC_PATH	CSV_PATH
5	2	./Output_Files/ASC	./Output_Files/CSV

“CONCEPTS.XML” is xml-formatted control file. If the xml-format is used, it must adhere strictly to the following –only one tag per line, with no spaces. The xml-formatted control file must be named “CONCEPTS.XML”. The following is an example of the xml-formatted control file showing all available keywords.

```
<TOPAGNPS>
<KEYWORD ID="ASC_PATH">./Output_Files/ASC</KEYWORD>
<KEYWORD ID="CSV_PATH">./Output_Files/CSV</KEYWORD>
<KEYWORD ID="DOWNSTREAM_REACH_ID">2</KEYWORD>
<KEYWORD ID="UPSTREAM_REACH_ID">5</KEYWORD>
</TOPAGNPS>
```

Keywords:

All parameters may be present but only required parameters and any desired optional parameters need be present. There are four possible keywords in the control file. All keywords must be spelled exactly as specified below and have a comma separating the keywords. The keywords are case and order insensitive.

“UPSTREAM_REACH_ID”:

This required keyword allows the user to specify the upstream end reach ID as the beginning point of the CONCEPTS corridor. This reach ID value must be a valid AnnAGNPS reach ID.

If this keyword's value is left blank then the default that will be used is the reach ID of the reach that contains the hydraulically most distant point to the watershed outlet.

“DOWNSTREAM_REACH_ID”:

This required keyword allows the user to specify the downstream end reach ID as the ending point of the CONCEPTS corridor. This reach ID value must be a valid AnnAGNPS reach ID.

If this keyword's value is left blank then the default that will be used is the reach ID of “2” which is the reach flowing into the watershed outlet.

“ASC_Path”:

This optional keyword allows the user to specify the path to where the spatial files created by DEDNM, RASPRO, RASFOR, and AgFlow are located. This path will also be used for spatial output files created by CONCEPTS.

This keyword is optional and if not included then the default location of the spatial input files and CONCEPTS spatial output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for all of the required spatial input files and CONCEPTS spatial output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for all of the spatial input files and CONCEPTS spatial output files.

NOTE: This keyword is only used for independent runs of the CONCEPTS module. It is ignored when CONCEPTS is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from CONCEPTS will be placed in the default folder location.

“CSV_Path”:

This optional keyword allows the user to specify the path to where the csv-formatted output files created by CONCEPTS will be written.

This keyword is optional and if not included then the default location of the CONCEPTS csv-formatted output files will be assumed. The default relative path is the folder where TopAGNPS was invoked. If the keyword is present with a path specified then that path will be used for all of the CONCEPTS csv-formatted output files. If the keyword is used but its value is blank, then the current path from which TopAGNPS is executed will be used for all of the CONCEPTS csv-formatted output files.

NOTE: This keyword is only used for independent runs of the CONCEPTS module. It is ignored when CONCEPTS is run in sequence with DEDNM, RASPRO, RASFOR, and AgFlow. In this case the output files from CONCEPTS will be placed in the default folder location.

Input data files:

If executing TopAGNPS on a UNIX/Linux system (case sensitive), the following list is in the expected case for each file.

The following input data files described below are spatial layers. The spatial layers must have the same dimensions with the same header information in each file. All of the filenames must be as specified below.

FLOVEC.ASC (TopAGNPS):

This required spatial input file is the flow vector layer.

AgFlow_Cell_Data.csv (AgFlow):

This required csv-formatted input file contains the cell information needed to produce the “CONCEPTS_Corridor_Cell-Reach_Coordinates.csv” output file.

AgFlow_Reach_Data.csv (AgFlow):

This required csv-formatted input file contains flow path information for the reach network.

AnnAGNPS_Cell_IDs.asc (AgFlow):

This required spatial input file is the subwatershed (AGNPS cell) designations layer.

AnnAGNPS_Reach_IDs.asc (AgFlow):

This required spatial input file is the reach (AGNPS reach) designations layer.

Output files:

There are currently seven output files generated by the program.

Log file:

The log file, “CONCEPTS.log”, is automatically generated and is a log of the processing as the program continues its execution. Additionally, the log files for “TopAGNPS” will also contain information specific for CONCEPTS. The log file, as it is being written, is also displayed to the screen. The log file contains processing information as well as error and warning messages.

Output data files:**AnnAGNPS_Output_Options_Data_Section_Cell_IDs.csv:**

This output file is a csv-formatted file that contains a list of the cell IDs within the CONCEPTS corridor that can be imported as the “Output Options - Cell” section into the AGNPS Input Editor or used directly by AnnAGNPS if used with the AnnAGNPS csv input option.

AnnAGNPS_Output_Options_Data_Section_Reach_IDs.csv:

This output file is a csv-formatted file that contains a list of the reaches within the CONCEPTS corridor that can be imported as the “Output Options - Reach” section into the AGNPS Input Editor or used directly by AnnAGNPS if used with the AnnAGNPS csv input option.

Concepts_Corridor_Cell_IDs.asc:

This is a computed spatial layer file containing AnnAGNPS cells within the CONCEPTS corridor.

Concepts_Corridor_Reach_IDs.asc:

This is a computed spatial layer file containing AnnAGNPS reaches within the CONCEPTS corridor.

Concepts_Non-Corridor_Cell_IDs.asc:

This is a computed spatial layer file containing AnnAGNPS cells that are outside of the CONCEPTS corridor.

Concepts_Non-Corridor_Reach_IDs.asc:

This is a computed spatial layer file containing AnnAGNPS reaches that are outside of the CONCEPTS corridor.

CONCEPTS_Corridor_Cell-Reach_Coordinates.csv:

This output file is a csv-formatted file that contains a list of the cell IDs within the CONCEPTS corridor and the coordinates to each cell's connection to the reach.

CONCEPTS_Corridor_Reach_Location_Coordinates.csv:

This output file is a csv-formatted file that contains a list of the reach IDs within the CONCEPTS corridor and the coordinates of the upstream and downstream ends of the reach.